

Robust Absolute Phase Unwrapping via Physics-Informed Convolutional Networks and Dynamic Noise Curricula

Ali Sattary

Department of Physics/Engineering, [University Name]

Abstract

Phase unwrapping is a mathematically ill-posed inverse problem critical to optical interferometry, synthetic aperture radar (SAR), and quantitative phase imaging. Classical methods, including branch-cut and minimum-norm algorithms, are notoriously fragile in the presence of intense optical noise and dense fringe discontinuities. In this proposal, we introduce a fully convolutional deep learning architecture designed for robust, direct absolute phase reconstruction from raw interferograms. We detail a modified U-Net architecture that explicitly integrates normalized spatial coordinate embeddings to resolve global scalar ambiguities. Furthermore, we propose a mathematically rigorous curriculum noise synthesis pipeline that progressively exposes the network to physically accurate optical aberrations, laser speckle, and quantum shot noise. By employing a multi-term loss topology encompassing point-wise absolute error, intensity-weighted Sobel gradient consistency, and discrete Laplacian curvature regularization, we hypothesize that the proposed model will significantly outperform classical approaches in noisy, real-world optical regimes.

I. INTRODUCTION

Optical interferometry relies on the superposition of electromagnetic wavefields to extract high-resolution topographical and structural data. However, sensors can only record scalar intensity. Consequently, the true continuous absolute physical phase $\phi(x, y)$ is recorded mathematically wrapped within the principal topological domain.

Let the wrapping operator $\mathcal{W} : \mathbb{R} \rightarrow [-\pi, \pi)$ be defined as:

$$\mathcal{W}(\phi(x, y)) = \phi(x, y) - 2\pi k(x, y) \quad (1)$$

where the topological shift field $k(x, y) \in \mathbb{Z}$ represents the unknown integer multiple of 2π at spatial coordinate (x, y) . Recovering the absolute continuous manifold $\phi(x, y)$ from the discrete observation $\mathcal{W}(\phi)$ is an ill-posed inverse problem due to the loss of $k(x, y)$.

Classical phase unwrapping techniques generally fall into two categories: path-following and minimum-norm approaches. Path-following methods attempt to evaluate the discrete gradient $\nabla \mathcal{W}(\phi)$, but struggle catastrophically when high-frequency optical noise triggers phase singularities, leading to unresolvable spatial error propagation. Minimum-norm approaches minimize derivative discrepancies via Poisson-like equations but fundamentally destroy the absolute macroscopic piston ($2\pi k$) structure.

Recent advancements in deep learning have demonstrated the capability of convolutional neural networks (CNNs) to implicitly map the nonlinear relationship between fringe intensity and absolute phase. In this research, we propose a physics-informed deep learning architecture designed to overcome the critical limitations of standard CNNs, specifically the $2\pi k$ translational ambiguity and the lack of physical noise resilience.

II. DETERMINISTIC SYNTHETIC DATA GENERATION AND PHYSICAL MODELING

A fundamental vulnerability of deep learning in physical sciences is domain shift caused by insufficient training diversity. To ensure robust generalization to empirical laboratory data while maintaining benchmark integrity, we developed a highly rigorous, deterministic synthetic data generator.

A. The Reproducibility Guarantee

To generate massive training distributions without compromising scientific benchmarking, the computational pipeline employs a master-seed architecture. A single root cryptographic seed deterministically branches into thousands of completely independent sub-seeds for concurrent multiprocess workers. This mathematically guarantees zero sample overlap or hash collisions across a 100,000+ image dataset, while ensuring the entire distribution can be perfectly regenerated by a third-party reviewer to validate performance benchmarks.

B. Forward Optical Model

The synthetic interferograms generated for this study model the interference of two coherent light beams. To standardize the topological geometry of the absolute phase fields, the optical generator strictly utilizes a fixed Helium-Neon laser wavelength of $\lambda = 632.8$ nm.

The primary wavefront $\phi_1(x, y)$ represents structural topography augmented with low-order polynomial aberrations. The reference beam $\phi_2(x, y)$ is mathematically modeled as a continuous divergent spherical wave originating from a longitudinal focal distance of $Z = 0.5$ m:

$$\phi_2(x, y) = \frac{2\pi}{\lambda} \sqrt{x^2 + y^2 + Z^2} \quad (2)$$

Let the base complex amplitude be denoted by E_0 . The recorded intensity field $I(x, y)$ represents the squared magnitude of the superposition of the two complex wavefields, resulting in the fundamental non-linear interferometry equation:

$$I(x, y) = \left| E_0 e^{i\phi_1(x, y)} + E_0 e^{i\phi_2(x, y)} \right|^2 = 2E_0^2 \left(1 + \cos(\Delta\phi(x, y)) \right) \quad (3)$$

where $\Delta\phi(x, y) = \phi_1(x, y) - \phi_2(x, y)$. This expansion rigorously demonstrates how the continuous physical phase difference is locked inside a periodic, non-invertible cosine manifold.

C. Supervised Regression Target

The objective of the model is to map the observed intensity $I(x, y)$ directly to the absolute phase difference between the two beams. To constrain the continuous scalar phase space, the supervised target $\Delta\phi(x, y)$ is shifted to a positive domain:

$$\Delta\phi(x, y) = (\phi_1 - \phi_2) - \min(\phi_1 - \phi_2) \quad (4)$$

III. PHYSICS-BASED DYNAMIC NOISE CURRICULUM

Neural networks are highly susceptible to domain shift when applied to empirical laboratory data. To close the gap between synthetic data and physical instrumentation, we introduce an extensive optical degradation pipeline mathematically anchored to the hardware physics of interferometric sensing.

A. Infinite GPU-Native Degradation

Classical deep learning datasets often compute noisy input pairs offline to mitigate input/output (I/O) bottlenecks. However, this permits the network to eventually memorize specific noise configurations. We circumvent this by porting the entire physical degradation model directly into the GPU batch-preparation thread. This guarantees that at every single epoch, the network encounters a mathematically unique, non-repeating noise topology for an identical underlying absolute phase structure, effectively acting as an infinite dataset.

The noise synthesis adheres strictly to the physical order of operations within an optical sensing array:

- 1) **Lens Aberration:** Optical defocus modeled as continuous Gaussian blur.
- 2) **Laser Speckle:** Multiplicative coherent noise: $I = I \cdot (1 + \mathcal{N}(0, \sigma_{\text{speckle}}))$.
- 3) **Uneven Illumination:** Low-frequency bicubic beam misalignment.
- 4) **Sensor Electronics:** Photometric gain and black-level offset jitter.
- 5) **Quantum Shot Noise:** Photon arrival statistics mathematically enforced via a positive-domain Poisson counting process. Crucially, this operation is executed prior to any non-physical data standardization (e.g., zero-mean Z-scoring) to preserve strict physical intensity boundaries.
- 6) **Thermal Noise:** Additive Gaussian white noise modeling sensor dark current.

B. Hybrid N-Cycle Stochastic Curriculum

Subjecting a neural network to peak optical degradation immediately at initialization typically causes gradient divergence. Conversely, training a network to convergence on a strictly static noise level severely narrows the breadth of the local minima it settles into.

To engineer maximum robustness, we introduce a *Hybrid N-Cycle Stochastic Curriculum*. Macroscopically, the overall noise level $\mu(\text{epoch})$ does not monotonically increase; instead, it forms continuous N-cycle cosine waves (mathematically equivalent to Warm Restarts). The noise gracefully rises to push the network to its structural limit, then drops back to near-zero, forcing the model out of sharp local minima and allowing it to fine-tune on perfect data.

Microscopically, the curriculum level is subjected to continuous Gaussian jitter $\gamma_{\text{batch}} \sim \mathcal{N}(\mu, \sigma_{\text{jitter}})$. This ensures that even at the macroscopic peak of the curriculum, the network is randomly forced to correctly reconstruct relatively clean interferograms, preventing catastrophic forgetting of the foundational physical mapping.

IV. NETWORK ARCHITECTURE AND SPATIAL ANCHORING

The backbone of the proposed method is a highly modified, physics-informed fully convolutional network (UNetRes2) specifically designed to resolve the macroscopic topological ambiguity of phase unwrapping while preserving high-frequency fringe structures.

A. Global Offset Head and Duality Mapping

The absolute physical phase represents a dual-scale topological space: high-frequency local fringe structures governed by continuous spatial gradients, and a low-frequency macroscopic piston shift ($2\pi k$). To mirror this physical duality, our architecture explicitly bifurcates the regression task. The standard U-Net decoding branch reconstructs the local topographical manifold $\phi_{\text{raw}}(x, y)$. Simultaneously, a parallel convolutional head stems from the deepest bottleneck layer and passes through Global Average Pooling to predict a singular scalar shift k_{off} . The final reconstructed absolute phase is assembled as $\phi_{\text{abs}}(x, y) = \phi_{\text{raw}}(x, y) + k_{\text{off}}$, preventing the network from distorting high-frequency local features to compensate for global scalar errors.

B. Depthwise Separable Res2 Blocks

Standard 3×3 convolutions possess insufficient receptive fields to track macroscopic phase trends across high-resolution 256×256 interferograms. To exponentially expand the spatial context without inducing catastrophic overfitting, the standard U-Net blocks are entirely replaced with Depthwise Separable Res2Net modules. Each block splits the channel dimension into $s = 4$ discrete groups and applies sequential, hierarchical depthwise convolutions with cumulative residual connections. This mathematically guarantees a massive effective receptive field capable of anchoring global phase structures while maintaining extreme parameter efficiency.

C. Multi-Scale Deep Supervision

Phase unwrapping networks frequently suffer from vanishing gradients and structural collapse due to the highly oscillatory nature of the input intensity field. To enforce structural coherence early in the training dynamics, the architecture utilizes Multi-Scale Deep Supervision. Auxiliary regression heads branch off the $1/4\times$ and $1/2\times$ scale decoding layers, returning explicit low-resolution phase predictions. Gradients are injected directly into the intermediate layers of the network during the backward pass, forcing the early decoding blocks to securely establish the macroscopic topological shape before fine-tuning the microscopic details at the terminal $1\times$ resolution.

D. Spatial Anchoring and Phase Hinting

Standard convolutions are translationally invariant, which fundamentally contradicts the spatially grounded nature of absolute phase unwrapping. Our architecture utilizes an `AddCoords` module, explicitly concatenating normalized spatial grid coordinates $(x, y) \in [-1, 1]^2$ to the input tensor to break the spatial invariance. Furthermore, the network is provided with a heavily regularized scalar hint ϕ_{hint} , corrupted by a random scalar perturbation δ , to force the network to actively align its predicted manifold with physical macroscopic anchors.

V. BAYESIAN HYPERPARAMETER OPTIMIZATION

Prior to the deployment of massive, long-duration training convergences, the architectural and topological hyperparameters are rigorously tuned via a fully automated Bayesian Hyperparameter Optimization (HPO) pipeline to escape suboptimal local minima.

A. Tree-Structured Parzen Estimator (TPE)

Because the hyperparameter search space (e.g., loss coefficient ratios, learning rate schedules) is highly non-linear, grid or random searches are computationally unviable. The pipeline utilizes the Tree-structured Parzen Estimator (TPE) algorithm. TPE constructs probabilistic models of the objective function, dynamically sampling configurations that maximize the expected improvement over the current best historical trial.

B. Median Pruning Dynamics

To mathematically conserve graphical processing unit (GPU) cycles, the tuning pipeline incorporates a Median Pruner mechanism. The performance of each active trial is evaluated at every epoch against the median historical Mean Absolute Error (MAE) trajectory of all previously completed trials. If a trial structurally underperforms the historical median at a specific epoch, it is aggressively and automatically terminated.

C. Optimization Space

The Bayesian optimizer actively explores a highly constrained parameter manifold to dictate the final training initialization. The search space explicitly tunes:

- 1) **Network Capacity:** The base channel width ([8, 16, 32]).
- 2) **Topological Loss Ratios:** The weighting coefficients for the spatial gradient constraint ($w_{\text{grad}} \in [0.01, 1.0]$) and Laplacian curvature penalty ($w_{\text{curv}} \in [0.0, 0.05]$).
- 3) **Convergence Dynamics:** The peak cyclic learning rate (log-uniform $[10^{-5}, 10^{-3}]$) and the initial warm-up sequence duration.

VI. MULTI-TERM TOPOLOGICAL LOSS

Due to the ill-posed nature of absolute phase unwrapping, optimizing the network weights θ exclusively via a point-wise absolute error typically fails to capture the high-frequency structural integrity of the continuous optical surface. Consequently, the network is optimized via a multi-objective loss topology $\mathcal{L}_{\text{total}}$ that rigorously enforces physical continuity, spatial derivatives, and localized confidence weighting:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MAE}} + \mathcal{L}_{\text{grad}} + \mathcal{L}_{\text{curv}} \quad (5)$$

A. Global Absolute Accuracy ($\mathcal{L}_{\text{MAE}}$)

The foundation of the optimization is the global L_1 norm (Mean Absolute Error), defined over the spatial dimensions $N = H \times W$. This term strictly anchors the unwrapped manifold to the absolute domain to eliminate the macroscopic $2\pi k$ piston shift:

$$\mathcal{L}_{\text{MAE}} = w_{\text{mae}} \frac{1}{N} \sum_{x,y} |\phi_{\text{pred}}(x, y) - \phi_{\text{gt}}(x, y)| \quad (6)$$

While $\mathcal{L}_{\text{MAE}}$ ensures convergence of the bulk surface geometry, it provides insufficient gradient signal to recover sharp, high-frequency topographical fringes, necessitating specialized structural regularization.

B. Intensity-Weighted Gradient Consistency ($\mathcal{L}_{\text{grad}}$)

To enforce high-fidelity structural reconstruction, we compute the spatial depthwise phase derivatives. We approximate the continuous gradients (∇_x, ∇_y) utilizing discrete 3×3 Sobel convolution kernels, denoted S_x and S_y :

$$\nabla_x \phi \approx S_x * \phi, \quad \nabla_y \phi \approx S_y * \phi \quad (7)$$

By penalizing discrepancies in the spatial derivatives, the network is strongly discouraged from generating overly smoothed, "averaged" phase fields that destroy sharp phase discontinuities.

Furthermore, we introduce a physical confidence weighting mechanism. In optical interferometry, regions of low intensity inherently exhibit low signal-to-noise ratios (SNR). Thus, we mathematically condition the gradient loss on the square root of the normalized raw intensity $I(x, y)$, effectively masking out derivative errors in dark regions while heavily penalizing structural topological deviations in regions of bright interference fringes:

$$\mathcal{L}_{\text{grad}} = w_{\text{grad}} \frac{1}{N} \sum_{x,y} \sqrt{I(x, y)} \cdot \left(|\nabla_x \phi_{\text{pred}} - \nabla_x \phi_{\text{gt}}| + |\nabla_y \phi_{\text{pred}} - \nabla_y \phi_{\text{gt}}| \right) \quad (8)$$

C. Laplacian Smoothness Regularization ($\mathcal{L}_{\text{curv}}$)

Convolutional decoding layers within U-Net architectures intrinsically generate high-frequency geometric artifacts (e.g., checkerboard artifacts) that violate the physical continuity of an electromagnetic wavefront. To enforce optical smoothness, we apply an L_1 penalty to the unrolled predicted surface utilizing a discrete 2D Laplacian operator.

Using the finite difference stencil formulation, the second-order spatial derivative at pixel coordinate (i, j) is computed as:

$$\nabla^2 \phi_{i,j} = \phi_{i+1,j} + \phi_{i-1,j} + \phi_{i,j+1} + \phi_{i,j-1} - 4\phi_{i,j} \quad (9)$$

The curvature penalty behaves akin to enforcing rigidity within an elastic membrane, effectively suppressing chaotic phase oscillations unconstrained by the first-order derivative loss:

$$\mathcal{L}_{\text{curv}} = w_{\text{curv}} \frac{1}{N} \sum_{i,j} |\nabla^2 \phi_{\text{pred}}(i, j)| \quad (10)$$

VII. EVALUATION METHODOLOGY

Evaluating absolute phase reconstruction is complex due to global scalar shifts (the "piston" effect). Before calculating final diagnostic metrics (RMSE, MAE, SSIM), the validation pipeline performs a Least-Squares Affine Alignment on the raw prediction:

$$a \cdot \phi_{\text{pred}} + c \approx \phi_{\text{gt}} \quad (11)$$

Metrics are computed on the aligned prediction $\hat{\phi} = a\phi_{\text{pred}} + c$. This explicitly isolates the network's topological unwrapping capability from trivial global drift errors.

VIII. RIGOROUS MLOPS EXECUTION PIPELINE

To ensure the absolute reproducibility and statistical integrity of the empirical results, the entire computational workflow is heavily decoupled into a strict, hardware-agnostic Machine Learning Operations (MLOps) pipeline. The execution sequence is formally enforced as follows:

A. Phase 1: Deterministic Generation (*generate*)

The pipeline is initialized by constructing the raw synthetic dataset. Because the topological generator is cryptographically seeded, the resulting HDF5 dataset structure is mathematically invariant across isolated computing environments, guaranteeing that identical noise configurations are presented to the network regardless of the underlying hardware.

B. Phase 2: Hyperparameter Optimization (*tune*)

Prior to full-scale training, the Bayesian TPE optimizer rigorously searches the non-linear configuration space (learning rates, loss topology weights, network capacity). To conserve compute limits, this search operates on a localized subset of the dataset. The optimizer ultimately isolates the absolute lowest-error configuration and explicitly freezes it into a `best_config.yaml` artifact.

C. Phase 3: Primary Training and Evaluation (*train*)

Utilizing the frozen configuration artifact, the architecture is trained from scratch to convergence on the massive primary dataset. Because the hyperparameters are structurally decoupled from the training runtime, the training thread guarantees zero experimental leakage. The model is subsequently evaluated against a completely sequestered, cryptographically isolated test set to extract raw diagnostic metrics (AbsMAE, TopoMAE, RMSE, SSIM, PSNR).

D. Phase 4: Statistical Validation (*multiseed and ablation*)

To defend against "lucky seed" anomalies, the final empirical metrics reported in the study are computed exclusively via an automated multi-seed aggregator. The pipeline sequentially spawns $N = 3$ completely independent training convergences utilizing the identical frozen configuration, each initialized with a vastly different network weight seed. The final TopoMAE and PSNR results are extracted by aggregating the individual test evaluations to formulate a mathematically unassailable Mean $\pm$ Standard Deviation. Finally, fully automated ablation sweeps isolate specific architectural mechanisms to mathematically quantify their explicit contribution to the structural resilience of the network. Specifically, the ablation suite evaluates the isolated removal of the Intensity-Weighted Gradient Loss (`no_grad_loss`), the Laplacian Smoothness Regularization (`no_curv_loss`), the Spatial Coordinate Embeddings (`no_coordconv`), and the Temporal EMA Shadowing (`no_ema`). By analyzing the resulting performance degradation, the empirical necessity of each theoretical mechanism is rigorously proven.

IX. RECENT IMPLEMENTATION REFINEMENTS

To strictly enforce the mathematical integrity and optical realism of the phase unwrapping pipeline, several core architectural optimizations were recently implemented:

A. Rigorous Optical Pipeline Ordering

The simulation of physical phenomena was reordered to faithfully match empirical laboratory conditions. Optical aberrations (lens defocus) now strictly precede multiplicative laser speckle, followed by macroscopic beam illumination biases, camera sensor clipping, quantum shot noise, and finally thermal Gaussian additive noise. This prevents physically impossible scenarios, such as optical blur being applied to internal electronic sensor noise.

B. Piston Hinting Consistency

We identified and corrected a mathematical contradiction in the phase piston anchor mechanism. Previously, the network was provided a scalar phase hint ϕ_{hint} corrupted by a random offset δ to prevent overfitting, but the regression target ϕ_{gt} lacked this matching translation. This penalized the network for successfully mapping the absolute depth scalar. By directly applying the perturbation δ to both the input hint and the ground truth target ($\phi_{\text{gt}} \rightarrow \phi_{\text{gt}} + \delta$), the network gradient directions are perfectly aligned with the absolute physical phase offset.

C. Intensity-Domain Quantum Shot Noise

The simulation of photon arrival statistics (Poisson noise) was restructured to operate exclusively within the strictly positive physical intensity domain. This resolved errors where standard normalization (Z-scoring) pushed intensity ranges into negative boundaries, causing non-physical zero-clipping and catastrophic mathematical degradation during subsequent noise augmentations.

D. Fractional Curriculum Scaling

The dynamic noise curriculum $\gamma(\text{epoch})$ was transitioned from absolute epoch tracking to relative fractional scaling (`warmup_ratio` and `full_ratio`). This ensures the curriculum function robustly self-scales regardless of the specific training horizon, enabling automated multi-trial hyperparameter optimization runs to perfectly compress and replicate the identical noise scheduling dynamics of massive long-duration convergences.

E. Temporal Weight Averaging (EMA Shadowing)

Because the stochastic optical noise curriculum is dynamically synthesized at every training iteration, the highly non-linear loss landscape constantly subjects the active network weights θ_t to chaotic, high-frequency gradient jitter. To isolate the stable structural minimum from this gradient noise, the architecture mathematically maintains an Exponential Moving Average (EMA) shadow model during the backward pass:

$$\theta_{\text{EMA}}^{(t)} = \beta \theta_{\text{EMA}}^{(t-1)} + (1 - \beta) \theta^{(t)} \quad (12)$$

with a rigid decay constant of $\beta = 0.999$. By evaluating validation metrics and extracting final predictions exclusively from the decoupled EMA weights, the pipeline acts as a temporal low-pass filter, mathematically guaranteeing that the final topological mapping represents the stable center of the convergence basin.

X. CONCLUSION

By rigidly marrying deep convolutional learning with established optical physics, the proposed pipeline represents a robust methodology for phase unwrapping. The implementation of spatial coordinate embeddings, dynamic optical noise scheduling, and multi-derivative loss topologies ensures that the network is explicitly trained to handle empirical, hardware-degraded interferograms with unprecedented accuracy.